coEdit

Secure Real-Time Collaborative Editing Platform

Backend Architecture & System Design

Presented by:

- Malak Mekyassi
- Marouane El-Hamdaoui

The Concept

The Challenge

Building a collaborative editor is deceptively complex. It requires handling concurrent edits from multiple users, resolving race conditions in milliseconds, and ensuring distributed state consistency without data loss.

The Solution

CoEdit uses a robust Spring Boot backend as the "Source of Truth." It manages state, enforces permissions, and broadcasts updates via WebSockets instantly to keep all clients in sync.

System Architecture

Client Layer

The frontend (React) connects via REST for management and WebSockets for live sessions.

App Layer

Spring Boot 3.5 handles business logic, security filtering, and real-time message routing (STOMP).

Data Layer

PostgreSQL ensures production-grade persistence, while JPA repositories manage efficient querying.

Real-Time Engine

The Real-Time Engine is the heartbeat of CoEdit. Using WebSockets, the system creates a persistent, two-way connection between the server and all active users. This architecture allows every keystroke, cursor movement, and presence update to be broadcasted instantly to all collaborators, ensuring everyone sees the exact same document state in milliseconds without ever needing to refresh the page.

Security & Access

JWT Auth

Stateless authentication. Every request carries a cryptographically signed token to verify identity.

Permissions

Granular RBAC ensures users only perform allowed actions: Owner, Editor, or Viewer.

Rate Limiting

Bucket4j implementation limits login attempts to 5 per minute to prevent brute-force attacks.

Data Integrity & Version History

Real-time speed must not compromise data safety. CoEdit implements a comprehensive versioning system that automatically snapshots documents during active sessions.

This creates a secure timeline of changes, allowing users to review past iterations and restore documents to any previous state.

Technology Stack

Core & Security

Spring Boot 3.5: The backend foundation.

Spring Security: Auth & Authorization.

JJWT: Token management.

Data & Utilities

PostgreSQL: Production Database.

Spring WebSocket: Real-time STOMP.

Bucket4j: Rate Limiting.

Live Demonstration

Auth Flow • Document Creation • Real-Time Sync